

# Overfitting Intuition

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can compare training and validation evidence for model capacity.

I can explain overfitting without assuming every gap proves it.

I can use a declared selection rule instead of perfect training fit.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

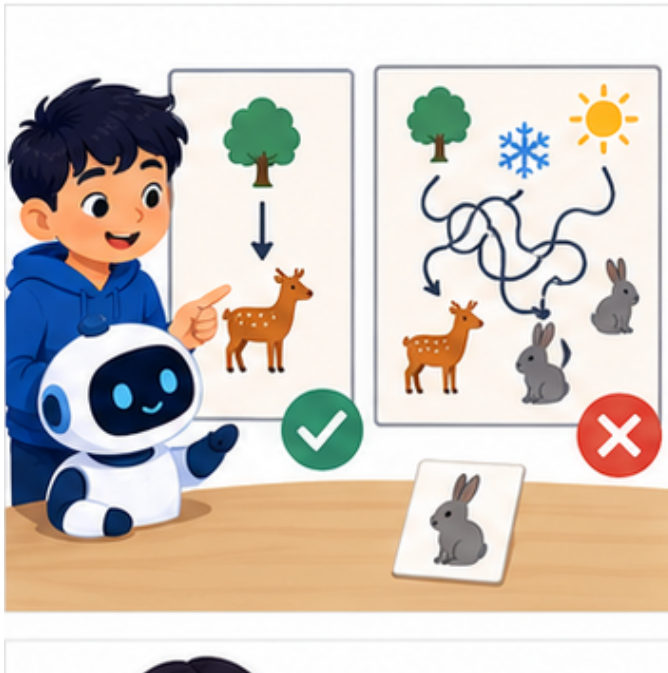
Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 16: Overfitting Intuition

A rule that memorizes examples may fail on new cases.



## THINKER MISSION

Contrast a simple rule with a seven-exception rule.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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## Readiness for the connected explanation

Can a zero training error guarantee good validation performance?

## This week's end goal

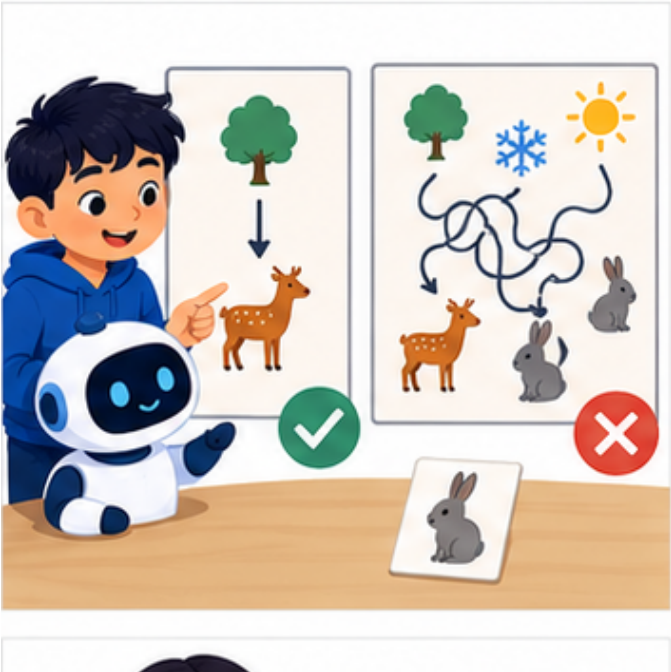
By the end, I can use development performance rather than training fit alone to choose model complexity.

# 2. Notice the evidence

## PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use overfitting intuition to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

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My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

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## A closer explanation

A more flexible model can represent complicated relationships and accidental details. Capacity can help when there is enough relevant evidence, but perfect training performance is not the goal. Compare training and validation using the same metric while checking leakage and representativeness.

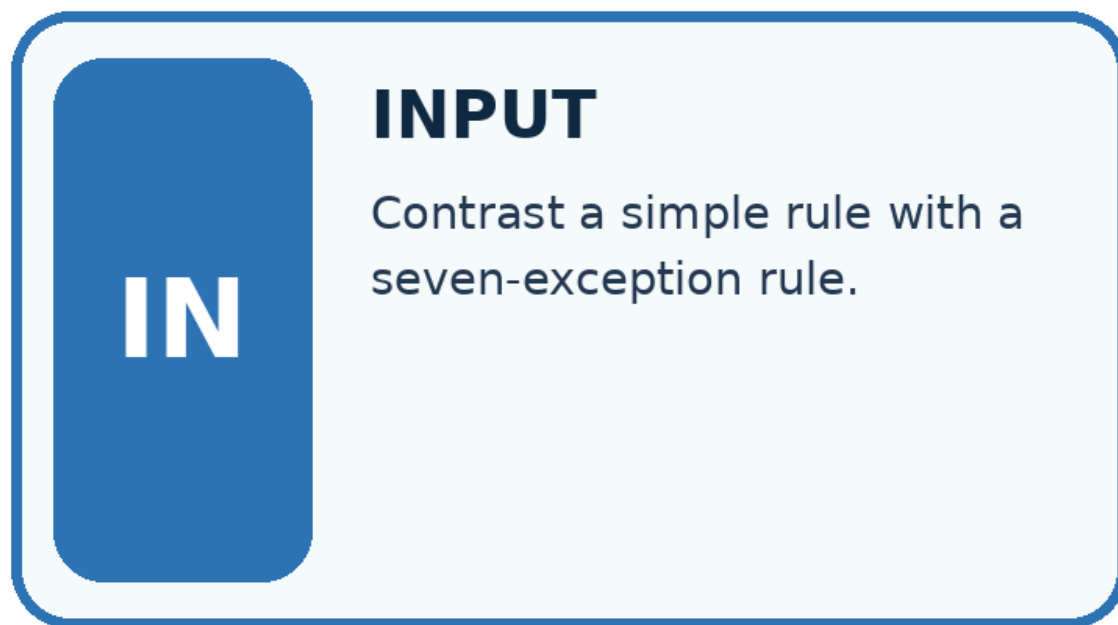
Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines overfitting and model capacity.

## 3. See how it works

### VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## PROCESS

A rule that memorizes examples may fail on new cases.



OUT

## OUTPUT

Contrast a simple rule with a seven-exception rule.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

On the same regression split, a constant has training MAE 5 and validation 6, a line has training 2 and validation 3, and a highly flexible curve has training 0 and validation 8. Under smallest validation MAE, choose the line. The curve has the lowest training loss but the worst validation result.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

## Connect the picture and the explanation

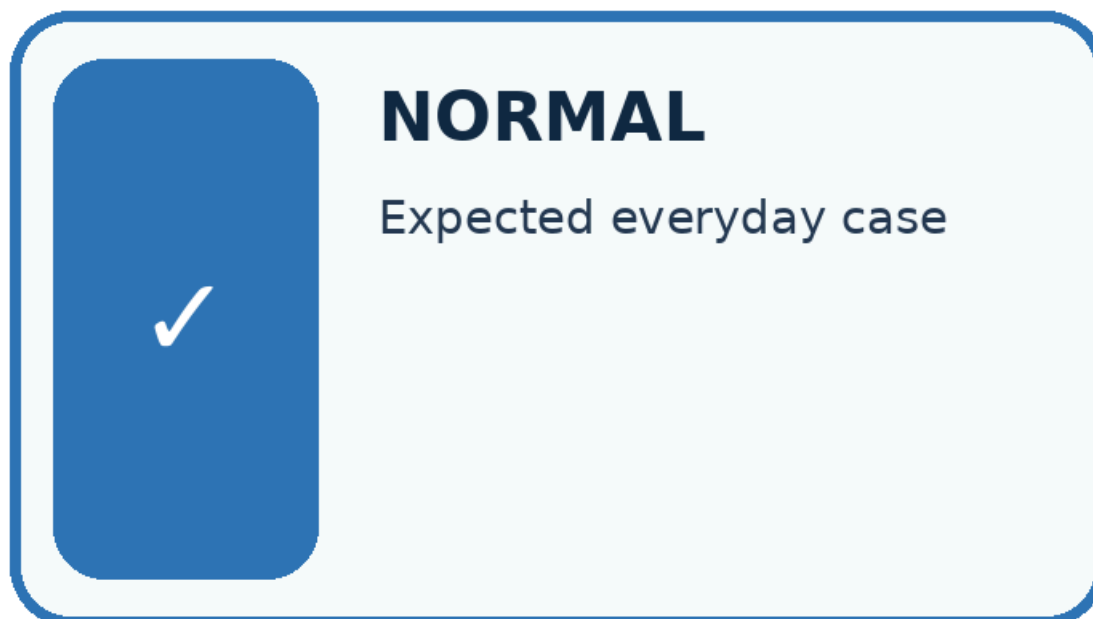
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





## BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: Complex's validation error becomes 2 on a newly documented, fair development comparison.

Prediction: The stated rule now selects Complex.

Reason: The rule compares validation errors; it does not ban complex models.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

**1**

## **OVERFITTING**

matching training details  
that do not generalize

**2**

## **GENERALIZE**

work on new relevant cases

3

## COMPLEXITY

number of conditions or exceptions

4

## EXCEPTION

special case outside the main rule

**Word · My own example · Not this / boundary**

**Word:** overfitting

**My own example:** Adapting too closely to development examples so performance does not transfer well to new relevant cases.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** generalize

**My own example:** Perform usefully on new relevant cases beyond those used to develop a system.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** complexity

**My own example:** The amount of structure, work or resources a method requires, measured in a stated way.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** exception

**My own example:** A condition that requires handling outside the usual path.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Capacity

**Meaning:** Flexibility of a model to represent patterns

\_\_\_\_\_

**Word:** Underfitting

**Meaning:** Failing to capture useful structure

\_\_\_\_\_

**Word:** Generalization

**Meaning:** Performance on relevant unseen examples

\_\_\_\_\_

## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

## WEEK 16 • WORKED EVIDENCE

- 1 Simple rule:  $\text{difficulty} \leq \text{reader level} + 1$ .
- 2 Memorized rule: seven title-specific exceptions.
- 3 Both fit the seen books; only new books reveal generalization.
- 4 Prefer the simplest rule that performs well on fair validation cases.

1. Simple rule:  $\text{difficulty} \leq \text{reader level} + 1$ .
2. Memorized rule: seven title-specific exceptions.
3. Both fit the seen books; only new books reveal generalization.
4. Prefer the simplest rule that performs well on fair validation cases.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

### Overfitting and model capacity at a glance

**Part · Evidence or action · Interpretation**

**Part:** Constant

**Evidence or action:** Training 5

**Interpretation:** Validation 6

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**Part:** Line

**Evidence or action:** Training 2

**Interpretation:** Validation 3

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**Part:** Flexible curve

**Evidence or action:** Training 0

**Interpretation:** Validation 8

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The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

**Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.**

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## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Overfitting Intuition

A fictional project compares three book-length prediction models on the same development sets.

### The facts we will use

Mean absolute errors in pages: Simple training 5, validation 6; Medium training 2, validation 3; Complex training 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

## Training and validation errors

|         | Training MAE pages | Validation MAE pages |
|---------|--------------------|----------------------|
| Simple  | 5                  | 6                    |
| Medium  | 2                  | 3                    |
| Complex | 0                  | 7                    |

Read a cell using both its row and column labels.

## Overfitting Intuition

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Training comparison

Complex fits training perfectly with zero error.



#### 2. Validation comparison

Medium has error 3, versus Simple 6 and Complex 7.



#### 3. Choose for development

Select Medium under the declared lowest-validation-error rule.



#### 4. Final evaluation

Freeze the selection, then evaluate new reserved cases.

### CHANGE ONE THING

Complex's validation error becomes 2 on a newly documented, fair development comparison.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Training comparison

**What happens and why:** Complex fits training perfectly with zero error.

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**Step:** 2. Validation comparison

**What happens and why:** Medium has error 3, versus Simple 6 and Complex 7.

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**Step: 3.** Choose for development

**What happens and why:** Select Medium under the declared lowest-validation-error rule.

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**Step: 4.** Final evaluation

**What happens and why:** Freeze the selection, then evaluate new reserved cases.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

A flexible model can fit quirks of its training set without predicting new examples well. Here Complex's training success occur together with the worst validation error. The numbers are invented comparisons, not a claim that complexity always hurts. Choosing Medium uses development evidence; its validation score must not be advertised as an untouched final-test result.

## What this example does not establish

A single small validation set may be noisy; selection evidence does not prove future population performance.

Read the labelled picture: Does zero training error justify choosing Complex in the original record?

Change one thing: Complex's validation error becomes 2 on a newly documented, fair development comparison. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: A rule that is perfect on training data must be the best rule. Strong move: locate evidence, change one cause, and retest.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

A simpler model is not always better. If a flexible candidate has better valid evidence under the declared rule, keep that evidence visible. Model capacity is one part of an experiment, alongside features, sample size, label quality and the split.

**What extra evidence would you inspect before concluding a model memorized noise?**

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## 8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on training and compare on validation. Do not remove difficult valid examples or repeatedly search the test for a favorable result.

## Practise with less help: Interpret a gap

Use workbook W2 for the connected example “Overfitting and model capacity”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Does zero training error justify choosing Complex in the original record?

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



### PREDICT → TEST → RECORD → EXPLAIN

Compare simple and exception-heavy rules on training cards, then on sealed new cards. Count where performance changes and explain why.

Compare simple and exception-heavy rules on training cards, then on sealed new cards. Count where performance changes and explain why.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

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**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



## MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
  - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
  - 3 • APPLY: Solve a different example independently.
  - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
  - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

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# 11. Connect your project

## CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



## TRANSPARENT BOOK RECOMMENDER

Avoid title-by-title patches; improve features or thresholds that transfer.

**THIS WEEK'S CHECKPOINT** Avoid title-by-title patches; improve features or thresholds that transfer.

**SYSTEM MAP** Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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**RESPONSIBLE DESIGN** Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

### Evidence for your project

Compare recommender revisions on fixed development cases, reserving new cases for the final

presentation.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



**INDEPENDENT PROOF**

Contrast a simple rule with a seven-exception rule.

CHALLENGE Contrast a simple rule with a seven-exception rule.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Rule A memorises six training book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

#### The question

What does the evidence support?

#### Worked explanation

A shows weak transfer. B might generalise better but requires testing; simplicity is not proof.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### Perfect recall can hide weak transfer

#### Evidence or stage · Value or result

**Evidence or stage:** Familiar books

**Value or result:** 6/6

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**Evidence or stage:** New books

**Value or result:** 2/6

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Rule A memorises six training book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

What does the evidence support?

A shows weak transfer. B might generalise better but requires testing; simplicity is not proof.

## 14. Try a new case independently

### Independent case: Apply independently

Models A training 4 validation 5, B training 1 validation 2, C training 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case

already practised with feedback cannot establish fresh transfer.

**Week 16 Independent application**

Close the worked explanation. Apply the idea to this different question without copying.

Rule A memorises six training book IDs. Rule B uses a relevant topic feature. Why might A fail on a new ID? What test is needed?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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**Create and test**

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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**Connect to your project**

Avoid title-by-title patches; improve features or thresholds that transfer.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

### Transfer practice from the original programme

Rule A memorises eight training IDs and gets 8/8 training, 2/8 validation. Rule B uses genre and gets 6/8 training, 6/8 validation. Which is preferable on this evidence, and what test remains?

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional project compares three book-length prediction models on the same development sets.

**By the end, I can · How I will show it**

**By the end, I can:** Compare training and validation evidence for model capacity

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Explain overfitting without assuming every gap proves it

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Use a declared selection rule instead of perfect training fit

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Compare training and validation evidence for model capacity

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional project compares three book-length prediction models on the same development sets.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Freeze the selection, then evaluate new reserved cases.

1. Final evaluation

2. Training comparison
3. Choose for development
4. Validation comparison

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Final evaluation Freeze the selection, then evaluate new reserved cases. A flexible model can fit quirks of its training set without predicting new examples well. Here Complex's training success occur together with the worst validation error. The numbers are invented comparisons, not a claim that complexity always hurts. Choosing Medium uses development evidence; its validation score must not be advertised as an untouched final-test result.

## Practice 2: explain the deciding evidence

Changed-case practice: Complex's validation error becomes 2 on a newly documented, fair development comparison. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The stated rule now selects Complex.

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The stated rule now selects Complex. The rule compares validation errors; it does not ban complex models.

## **Repair a missing building block**

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf predictions from training examples. Short practice: Choose the best original question and explain why its more balanced split matters here.. Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?. Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated input format, selected rule and recorded output. Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

## **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.